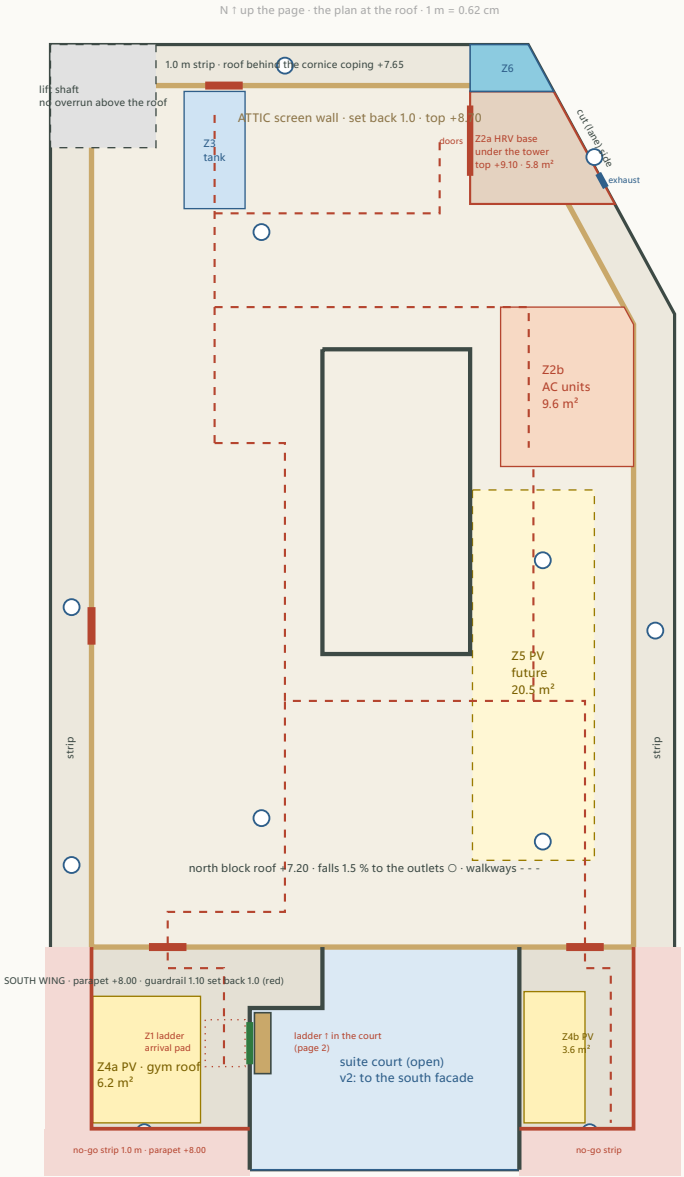
Roof study · the maintenance ladder, the lowered attic, the low parapet with the set-back guardrail, the main cornice, the reserved zones · Taif / Palm House V8.1

Design review F-07 (major) · owner decisions 31 Oct 2026, V8.0 and V8.1 (the maintenance ladder, the roofline lowered) · flat roof, no tiles · the roof for maintenance and equipment only · no slab level changes · all sizes assumed until the engineer, the ladder supplier and the fall-protection specialist confirm



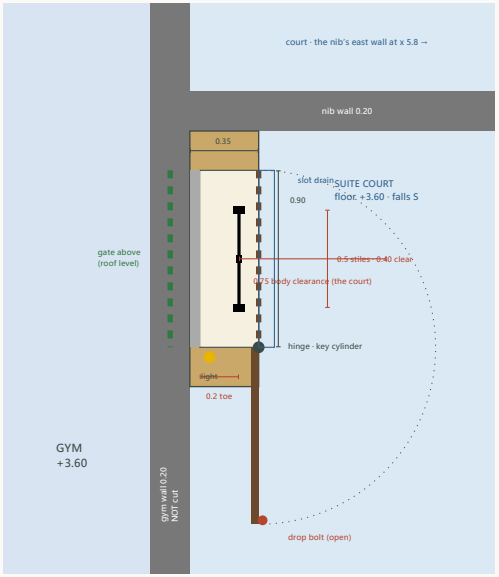
1 · Roof plan: what is on the roof

- **Access (owner, V8.1):** the roof is reached by a **fixed maintenance ladder Z1 in the suite court** (page 2): 3.7 m from the court floor +3.60 to the gym roof +7.30, in a recess in a stone pilaster on the gym's east wall, behind a locked panel; a self-closing gate in the court's west guard at the top. Workers reach it through the house (gym → court). The main stair stops at the first floor and the roof slab over its well is closed; no door from the roof into the house. The lift stops at the first floor; its shaft needs no head above the roof (§ 5).
- **Attic (lowered):** a screen wall 1.4 m above the roof finish, coping included (top +8.70 was 1.80 / +9.10), set back 1 m from the facade face on the north, the cut (lane), the east and the west sides (a plain pentagon); its south wall stands on the y 19.25 line (beams B8 / R5). Length ≈ 57.8 m. Two openings to the strip (north, west; gated and locked) and two to the south wing roofs (x 2.1–2.9 and x 11.0–11.8).
- **The 1.0 m strip** (Z7) in front of the attic: roof behind the main cornice and its low coping, with its own outlets; a no-go zone (harness).
- **South wing** (y 19.25–24): the main cornice and a **low visible parapet** 0.7 m above the finish (top +8.00 was 1.10 / +8.40); the fall protection is a **continuous 1.10 m black steel guardrail set back 1 m** from the west, south and east faces (red), running from the attic's end walls to the court guards: not seen from the street; the strip between it and the parapet (Z8) is a no-go zone. The roofline steps 0.7 m down from the attic coping.
- **Guards:** a black steel bar guard 1.10 m round the courtyard sky hole (19.3 m) and on the suite court's west (gym roof and nib) and east (office roof) edges; its north edge is the attic wall, its south edge the parapet on the facade frame R8. The guarding is **continuous from the ladder's arrival to every piece of equipment** (checked, § 3): the gym roof zone, the attic (its wall is the guard), the office roof zone.
- **Drains:** 11 roof outlets to internal downpipes in the walls (no scuppers through the cornice); falls 1.5 % away from the holes and the attic walls; the strip falls along the facade to its outlets; overflow spouts in the strip's upstand.
- **Walkways:** 0.9 m wide on paving pads from the ladder's arrival pad north (x 3.25–4.15, between the PV and the court guard) through the attic opening at x 2.1–2.9 into the north block, then the loop to the tank (Z3), the HRV base's doors (Z2a), the condensers (Z2b) and the PV (Z5); a branch through the attic opening at x 11.0–11.8 along the office roof (x 11.5–12.4) to the PV Z4b; no opening crossed, nothing in the no-go strips, no walking on the insulation.

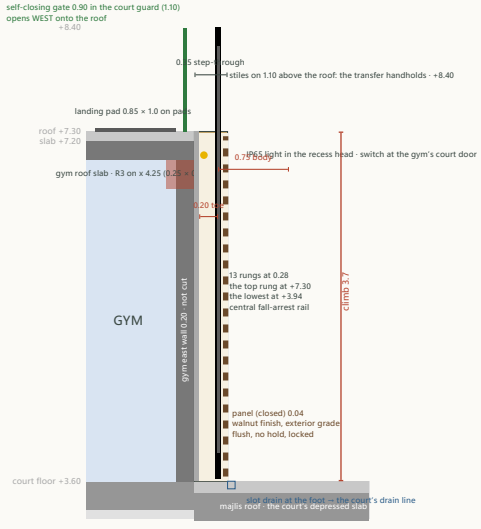
Zone	m²	Note
Z1 the maintenance ladder (suite court → gym roof)	0.5	V8.1 (owner): the roof is reached by a fixed vertical ladder in a recess in a pilaster on the gym's east wall (page 2): 3.7 m climb from the court floor +3.60 to the gym roof +7.30, behind a locked walnut-finish panel; the internal stair stops at the first floor, the roof slab over the well closed
Z2a the HRV base under the tower	5.8	`studies/air_system.typ` v3.1 (owner): the tower rises from a base housing the ventilation unit (one mass): top +9.10 (0.4 m above the lowered attic coping: part of the malqaf mass on the facade, seen by design; § 8), 1.65 m clear inside (a low service compartment: the unit ≈ 110 kg on a plinth right behind a pair of doors 1.5 m high on its west face, filters and the core pulled from the roof terrace, no entry); the intake duct drops inside the tower's compartment; the exhaust louvre on the cut face
Z2b plant: AC outdoor units	9.6	behind the cut / east attic wall, SOUTH of the tower and its base (owner): 2.2 m clear of the base's south wall, 5.15 m south of the intake louvre in plan, so the tower never takes their hot air; over the suite bath (anti-vibration mounts); 0.6 m service space round each unit
Z3 water tank(s)	3.3	≈ 2 × 1.5 m³ between the roof beams R10 and R3, inside the attic (needed only if the owner wants a gravity reserve)
Z4 PV, south wing (gym roof Z4a + office roof Z4b)	9.8	west and east of the court, never in front of the clerestory windows; V8.1 only inside the set-back guardrail and off the walkways (was 24.6 m²: the no-go strips and the ladder's walkway take the rest); fixing points cast in before waterproofing
Z5 PV, future (north block east)	20.5	inside the attic; morning shade from the east attic wall
Z6 malqaf head	1.5	the tower rises to +9.80: with its base the one element above the attic, by design; v3: a four-sided head, the breeze shaft and the ventilation intake compartment inside (`studies/air_system.typ`)
Z7 the 1.0 m strip in front of the attic	40.6	roof behind the main cornice and its coping; falls to its own outlets; a NO-GO zone (the coping is 0.35 above the finish): its two openings in the attic wall get locked gates; entered only on a harness clipped to anchor eyes on the attic wall
Z8 the no-go strips of the south wing	13.6	the 1.0 m between the guardrail and the parapet (+8.00, 0.70 above the finish): no equipment, no walkway; the outlets there reached on a harness

Areas on wall centre lines / zone outlines. PV: ≈ 3.1 + 10.3 panels of 2 m² (≈ 5.6 kWp at 420 W each) if both zones are used.

The roof access · the maintenance ladder in the suite court (owner, V8.1) · enlarged plan and section · the main stair stops at the first floor



PLAN at the court floor · 1 m = 2.6 cm · north up

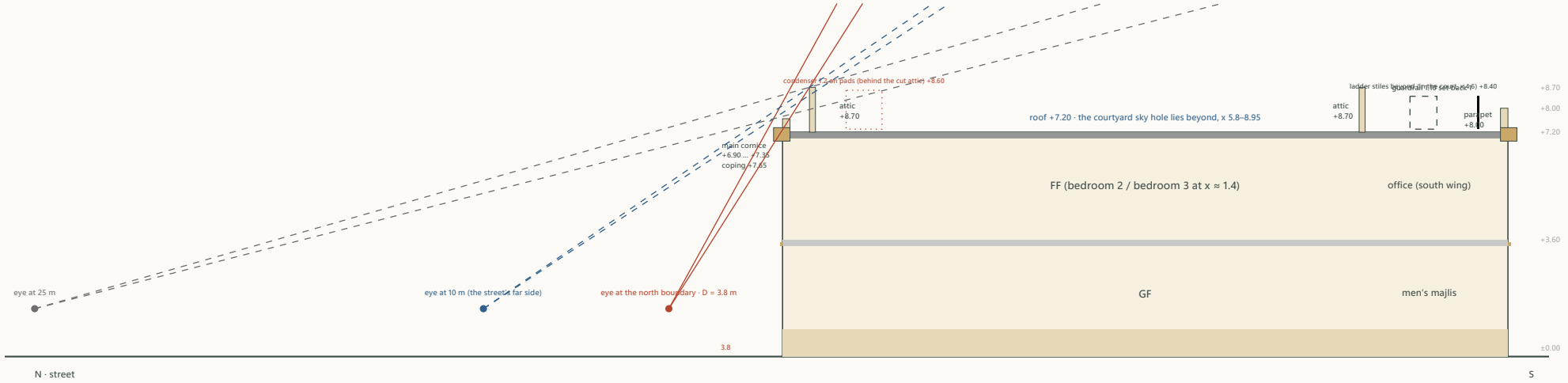


SECTION E–W through the recess (y 21.3), looking north · 1 m = 1.25 cm · W ← → E

2 · The roof access: the maintenance ladder

- **Owner's decision (V8.1):** a simple roof access (cost and appearance); office visitors are met in the men's majlis, the office is the owner's only, reached from the suite court. The roof is reached by a **manufactured vertical fixed ladder** (not site-made; black matte, non-slip rungs, corrosion-protected, aluminium or hot-dip galvanised and powder-coated steel) in the suite court, on the **gym's east wall**, at the north end of the wall against the nib: away from the seating (y ≥ 21.7) and off the door strip (y ≤ 20.55, x ≥ 5.9).
- **The recess:** 0.9 wide × 0.3 deep, full height from the court floor +3.60 to the gym roof +7.30, formed in a **honey-stone pilaster added on the court face of the wall** (1.3 × 0.35 m, cheeks 0.20): the 0.20 m structural wall on B19 (under R3) is **not cut**. If the engineer confirms that wall as non-load-bearing infill between B19 and R3, the recess may instead be cut into a locally rebuilt 0.50 wall and the pilaster disappears (structural coordination, X20). The pilaster's own weight (≈ 1.7 m³ block ≈ 20 kN) stands on the court's depressed slab: the engineer.
- **The ladder:** stiles 0.5 overall, rungs 0.40 clear, pitch 0.28, 13 rungs, the **top rung level with the roof finish** (+7.30), the lowest 0.34 above the floor; **toe clearance 0.2 m** behind the rung line (≥ 0.18), **body clearance ≥ 0.75 m** in front of it (the court, with the panel open); climb **3.7 m**. The stiles run on **1.10 m above the roof** (+8.40) as the transfer handholds (fixed, so no retractable grab post; if one is preferred it is mechanically fixed, never friction-held). Tools go up in a bag: both hands on the ladder.
- **The panel:** exterior-grade walnut-finish (marine ply or aluminium composite with a walnut veneer, 0.04 thick), hinged on the south jamb, **opens 180°** to lie flat along the pilaster line (x 4.7, y 21.75–22.65), held open by a **floor drop-bolt** away from the climber; a key cylinder outside (the key with the parents / the maintenance key); **closed it is flush with the pilaster face and offers no hold**: not climbable (children). With the panel closed the court reads a walnut panel in a stone pilaster, in the design language.
- **The arrival on the roof:** the climber steps from the top rung over the pilaster's top (0.35) through a **self-closing gate 0.90 in the court's west guard** (1.10 m, on x 4.25, directly above the recess), opening **west onto the gym roof**, onto a **standing landing pad** (x 3.3–4.15, y 20.8–21.8) inside the protected zone; from it the walkway north to the attic opening. The guarding is continuous from the pad to every piece of equipment (§ 3 check); the strip outside the set-back guardrail is a no-go zone.
- **Fall protection:** the climb (3.7 m) is over the 3.0 m threshold of EN ISO 14122-4: the ladder is specified with a **central fall-arrest rail** (EN 353-1 glider) rather than a cage, which would not fit the recess; the specialist confirms the arrangement under the Saudi requirements (SBC / Civil Defense / the labour safety rules) — **not verified here**. Anchor eyes on the attic wall for the no-go strips.
- **Services:** an IP65 light in the recess head with the switch at the gym's court door (and the court's lights); the recess floor falls 1 % out to a **slot drain** at the pilaster's foot tied into the court's drain line (`studies/ff_suite.typ`); no dust pocket, no standing water.
- **The internal stair stops at the first floor** (V8.0): the roof slab over the well is an ordinary bay (no R16–R18), no stair room; the attic a plain pentagon. The lift: no change (§ 5). **3D (V8.1):** the pilaster, the closed panel, the stiles, the gate and the set-back guardrail are built from `3d/params.yaml` `roof_design.ladder` / `guardrail_setback`.

Sight lines · north–south section at x ≈ 1.4, looking west · the attic 1.4 m above the finish, set back 1 m · the parapet 0.7 m and the set-back guardrail · eye 1.6 m above the ground



3 · What the attic hides, and from where

The eye is 1.6 m above the ground outside (+1.30 against the house's ±0.00; the ground assumed at −0.30). An object behind the facade is hidden when its top lies below the line from the eye over the screen in front of it (the cornice coping +7.65 on the facade, or the attic coping +8.70 1 m behind it).

Observer (north street)	cornice	attic	condensers (1.2 m, 2.0 m behind the facade)
north boundary (street), NW corner	59.1°	57° (hidden)	51.5° hidden
north street, 10 m from the facade	32.4°	33.9°	31.3° hidden
north street, 25 m (the far side)	14.3°	15.9°	15.1° hidden

Angles above the horizon from the eye. From the lane boundary (D = 2.5 m) the condensers (1.2 m on 0.10 pads, 2.0 m behind the cut facade: top +8.60) are seen at 58.3° against the attic's 64.7°: hidden; and from every distance, since +8.60 ≤ +8.70. The tank (top +8.30) likewise. From the east boundary (D = 3.6 m) the plant zone is likewise hidden. The malqaf head (+9.80) and its HRV base (+9.10), on the facade in the cut tip, are the one element above the attic, by design.

- **Attic height (owner, V8.1): 1.4 m above the roof finish, coping included** (top +8.70 was 1.80 / +9.10). **Re-run:** the minimum that hides the condensers from the street and the lane at eye level from every distance is 1.3 m (1.2 m units on 0.10 pads: top +8.60); 1.4 m leaves 0.10 m of margin, so **1.40 m works** on the condition that the outdoor units are **≤ 1.2 m high** (most 2–7 kW split and VRF condensers are 0.6–1.0 m; a tall unit laid on its side is not allowed: specify the height to the MEP engineer). The tank 1.0 m: hidden with 0.40 m of margin.
- **The HRV base** (+9.10) now stands 0.4 m above the attic coping: it is on the facade line in the cut tip, the plinth of the malqaf mass (seen by design, like the head). **Option** (owner / MEP): lower it to +8.70 with 1.25 m clear inside and 1.1 m doors (a horizontal unit ≤ 0.7 m); drawn: kept at +9.10 (1.65 m clear). The ladder's stiles (+8.40, in the court 2.4 m behind the court's south face) stay below the attic coping and are hidden from the south garden by the parapet up to ≈ 40 m; black matte.

4 · The courtyard's sky and sun with the attic

From every point of the courtyard floor the attic walls lie **below the hole's own edges** (the roof edge beams at +7.20), so they take nothing from the sky the courtyard sees today (enclosure study: 7.2 % of the hemisphere from the floor's middle, unchanged):

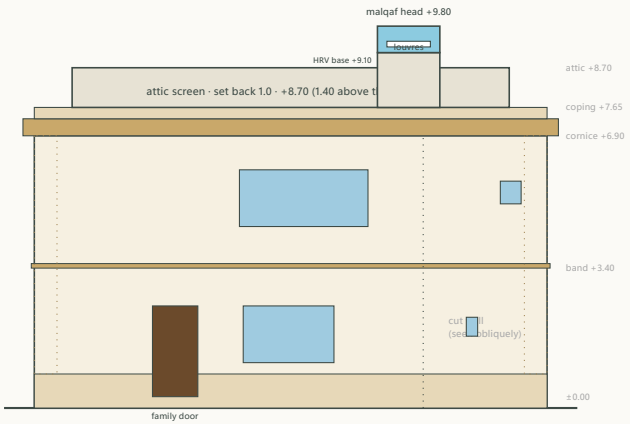
From the far edge of the floor	hole edge	attic
attic south wall (y 19.25, +8.70) from the floor's north edge (y 6.6)	48.4°	34.5° ✓
attic north wall (y 0.875) from the floor's south edge (y 12.9)	48.4°	35.9° ✓
attic east wall (x 12.436) from the floor's west edge (x 4.35)	58°	47.1° ✓

- **Winter noon sun** (45.2°, the light check): the ray that just clears the hole's south edge (+7.20 at y 13.0) passes the attic's south wall line (y 19.25) at +13.49, far above its coping +8.70: the attic never shades the courtyard, in any season. The clerestory windows sit in the y 19.25 wall itself, below the attic wall: unchanged; their sun comes over the court's south frame, whose parapet is now +8.00 (0.40 lower: a little more winter sun, `studies/sections.typ`).
- **The suite court** gets the attic's south wall on its north side (its shaded side): no loss of sun, a little less sky.

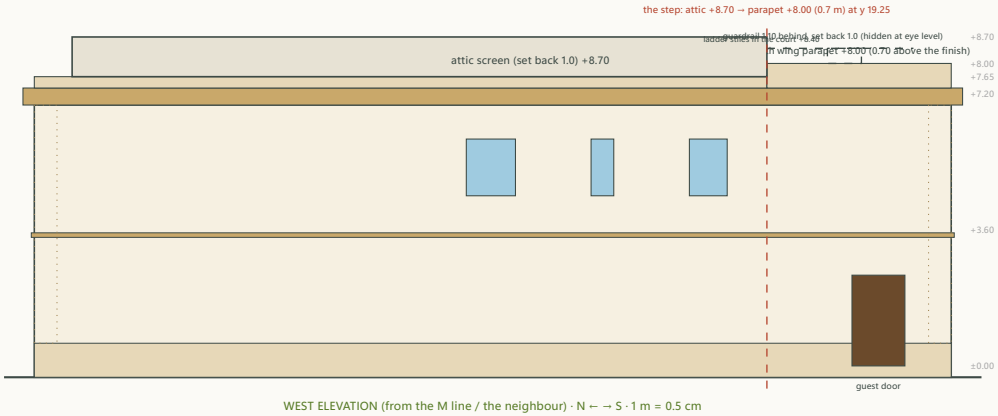
5 · The lift

- Top stop at the first floor (+3.60); the roof slab underside at +7.00: **3.4 m headroom available**. A typical Saudi villa lift (machine-room-less traction, 400–630 kg, 2 stops) asks 3.4–3.6 m headroom above the top floor and a 1.2–1.4 m pit (assumed from supplier catalogues; open item 14 asks for the drawings); a platform home lift (EN 81-41) asks 2.3–2.6 m.
- **So:** choose a lift with headroom ≤ 3.4 m: no shaft head above the roof. If the chosen lift needs up to 3.85 m, a raised shaft head of ≤ 0.45 m (top +7.65) stays hidden behind the main cornice coping; more than that is not possible: the shaft (x 0–2.25, y 0–2.2) lies in the 1.0 m strip in front of the attic, not inside it. Added to open item 15 as a requirement for the supplier.

North and west elevations · simple line drawings · the main cornice, the thin cornice between the floors, the lowered attic, the low parapet, the stepped roofline · openings indicative



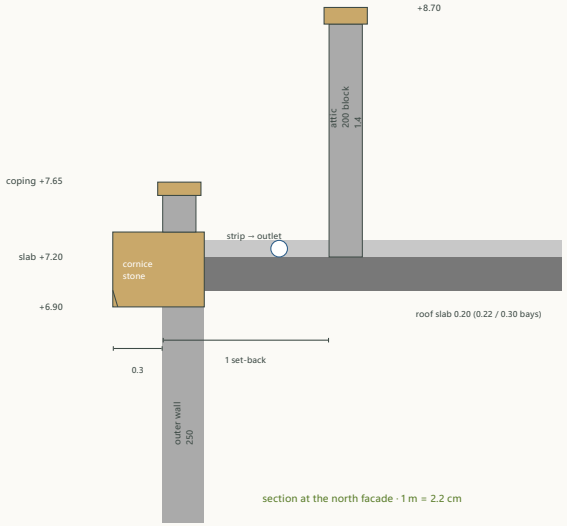
NORTH ELEVATION (from the street) · W ← → E · 1 m = 0.5 cm



WEST ELEVATION (from the M line / the neighbour) · N ← → S · 1 m = 0.5 cm

Cream plaster, honey stone at the plinth (to +0.60), the corners (quoins, dotted), the thin band at the first-floor slab (0.12 m, projecting 0.08) and the main cornice (0.45 m high, projecting 0.3, +6.90 … +7.35); the attic plastered cream with a stone coping. The south elevation repeats the west one's south part (parapet +8.00, the court's screened opening under the frame); the east elevation the north block's coping and attic (set back 1.0) over the kitchen / suite windows, then the parapet (the east elevation is drawn below). Openings are indicative here: the real ones are in the plans and the studies.

The cornice and the attic in section · loads for the engineer · what changes in the files



section at the north facade · 1 m = 2.2 cm

6 · Sizes (assumed, by this study)

- **Main cornice:** honey stone, 0.45 m high (+6.90 … +7.35), projecting 0.3 m with a drip, on the facade edge all round the house; above it a 0.30 m upstand with a stone coping to **+7.65** on the north block (the strip's parapet) and the plain parapet to **+8.00** on the south wing. The thin cornice between the floors: a stone band 0.12 m high at +3.40, projecting 0.08.
- **Attic (lowered V8.1):** 200 mm block, plastered cream, stone coping; 1.4 m above the roof slab + 0.10 coping = 1.4 m above the finish (coping +8.70); set back 1 m from the facade face (0.88 m from the wall centre lines); ≈ 57.8 m long; on the south on the y 19.25 line (B8 / R5). Four openings 0.80 (two to the strip, gated and locked; two to the south wing roofs).
- **South wing parapet:** 0.7 m above the finish, coping included (top +8.00); the **guardrail** 1.10 m black steel bars (posts at 1.5 m) set back 1 m from the outer faces, continuous from the attic's end walls to the court guards; the no-go strips 1.0 m.
- **Roofline step:** 0.7 m at y 19.25 (attic coping +8.70 → parapet +8.00), plus the 1.0 m plan set-back. **Final heights:** cornice +6.90 … +7.35, north coping +7.65, south parapet +8.00, attic coping +8.70, HRV base +9.10, malqaf head +9.80, ladder stiles +8.40 (inside the court). Targets, to confirm at the facade design stage.
- **Maintenance ladder** (page 2): recess 0.9 × 0.3 in a stone pilaster 1.3 × 0.35 on the gym's east wall; climb 3.7 stiles to +8.40 the gate 0.90 and the pad on the gym roof.

7 · Loads for the engineer (assumed)

- **Attic walls:** 200 block 1.4 m + coping ≈ 3.8 kN/m (was 4.5), standing on the roof slab 0.88 m inside the outer-wall beams (R7 / R8, the cut and east wall beams): near the supports, but a line load on the slab: thicken the slab edge or add a small hidden beam under it; the south attic wall on R5 / B8.
- **Parapet and guardrail (south wing):** the 0.70 parapet ≈ 1.8 kN/m on the facade beams (was 2.8); the 1.10 m guardrail on the slab 1.0 m in: posts at 1.5 m on base plates bolted to the slab through the waterproofing (or on a 0.20 kerb), designed for a 0.74 kN/m horizontal load at the top rail and 0.9 kN point load (SBC 301 / ASCE 7 guards): small, but the anchors and the upstand detail are the engineer's.
- **Ladder and pilaster:** the pilaster ≈ 20 kN on the court's depressed slab (the men's majlis roof); the ladder's fixings into the pilaster (resin anchors into solid block or a cast-in steel sub-frame): the supplier's loads (≈ 1.5 kN per fixing); the gate posts on the roof edge beam R3.
- **Cornice:** stone ≈ 2.5 kN/m hung on the slab edge (anchored precast or cut stone); the thin band 0.3 kN/m.
- **Plant:** 3–4 AC outdoor units ≈ 1.0 kN each and the ERV ≈ 1.5 kN on pads with anti-vibration mounts (local ≈ 1.5 kN/m²); **tank(s):** up to 3 m³ = 30 kN on ≈ 3.3 m² (≈ 9 kN/m²) on the R10–R3 bay (1.5 m span): to be checked; **PV:** ≈ 0.25 kN/m² plus wind uplift (ballast or cast-in fixing points, 24 m² now, 20 m² future); **malqaf head** as the air study; walkway pads 0.5 kN/m².
- **Roof live load** for maintenance only: 1.0–1.5 kN/m² (SBC 301, the engineer).

8 · Conflicts and open points

- **The HRV base vs the lowered attic** (§ 3): the base (+9.10) is 0.4 m above the attic coping; kept as part of the malqaf mass (seen by design) or lowered to +8.70 with 1.25 m clear inside: owner / MEP.
- **Workers through the house:** roof maintenance enters by the gym and the suite court (the family's private side); the ladder's key is a household key. Accepted by the decision (V8.1). The site has one entry: the farm's 4 m main gate (no gate on the lane).
- **PV:** the no-go strips and the ladder's walkway cut the south wing PV to 9.8 m² (was 24.6); Z5 on the north block (20.5 m²) is unchanged: R-31 to revisit with the site plan.
- **Lift** (§ 5): no overrun if the supplier's headroom is ≤ 3.4 m (≤ 3.85 m with a 0.30 m raised head behind the cornice): a requirement for open item 15.
- **Courtyard light** (§ 4): none lost. **First-floor layout:** the office has one door (from the court) and an east window; the suite bedroom's escape window runs y 15.0–18.6 (`studies/ff_suite.typ`).
- **3D, DXF and the presentation (V8.1):** the ladder, the 1.40 attic, the 0.70 parapet and the set-back guardrail are built and drawn.

9 · What this study changes in the files

- `studies/structure.typ`: R16–R18 and the stair-well voids dropped (V8.0), X16 superseded; X20 the ladder recess / pilaster, X17 the lowered attic, the 0.70 parapet and the guardrail loads; `studies/fire_escape.typ`: the first floor's exits are the main stair and the escape windows, the court a fresh-air area, not an exit; `3d/params.yaml`: the attic / parapet / ladder / guardrail data (V8.1).
- `docs/DESIGN_LANGUAGE.md`, `docs/OPEN_ITEMS.md`, `PROJECT_V8.md` (V8.1).
- `data/decisions.csv`: the roof by a maintenance ladder, the roofline lowered (V8.1).

Study checks passed (18)